Solutions – Problem set on Resonance

1.

a. $f_s = \dfrac{1}{2\pi\sqrt{LC}} \Rightarrow L = \dfrac{1}{(2\pi f_s)^2 C} = \dfrac{1}{(2\pi\ 1.8\text{ kHz})^2\, 2\ \mu\text{F}} = \mathbf{3.91\ mH}$

b. $X_L = 2\pi fL = 2\pi(1.8\text{ kHz})(3.91\text{ mH}) = \mathbf{44.2\ \Omega}$

$X_C = \dfrac{1}{2\pi fC} = \dfrac{1}{2\pi(1.8\text{ kHz})(2\ \mu\text{F})} = \mathbf{44.2\ \Omega}$

$X_L = X_C$

c. $E_{\text{rms}} = (0.707)(20\text{ mV}) = 14.14\text{ mV}$

$I_{\text{rms}} = \dfrac{E_{\text{rms}}}{R} = \dfrac{14.14\text{ mV}}{4.7\ \Omega} = \mathbf{3.01\ mA}$

d. $P = I^2R = (3.01\text{ mA})^2\ 4.7\ \Omega = \mathbf{42.58\ \mu W}$

e. $S_T = P_T = \mathbf{42.58\ \mu VA}$

f. $F_p = \mathbf{1}$

g. $Q_s = \dfrac{X_L}{R} = \dfrac{44.2\ \Omega}{4.7\ \Omega} = \mathbf{9.4}$

$BW = \dfrac{f_s}{Q_s} = \dfrac{1.8\text{ kHz}}{9.4} = \mathbf{191.49\ Hz}$

h. $f_2 = \dfrac{1}{2\pi}\left[\dfrac{R}{2L} + \dfrac{1}{2}\sqrt{\left(\dfrac{R}{L}\right)^2 + \dfrac{4}{LC}}\right]$

$= \dfrac{1}{2\pi}\left[\dfrac{4.7\ \Omega}{2(3.91\text{ mH})} + \dfrac{1}{2}\sqrt{\left(\dfrac{4.7\ \Omega}{3.91\text{ mH}}\right)^2 + \dfrac{4}{(3.91\text{ mH})(2\ \mu\text{F})}}\right]$

$= \dfrac{1}{2\pi}\left[601.02 + 11.324\times10^3\right]$

$= \mathbf{1897.93\ Hz}$

$f_1 = \dfrac{1}{2\pi}\left[-\dfrac{R}{2L} + \dfrac{1}{2}\sqrt{\left(\dfrac{R}{L}\right)^2 + \dfrac{4}{LC}}\right]$

$= \dfrac{1}{2\pi}\left[-601.02 + 11.324\times10^3\right]$

$= \mathbf{1.71\ kHz}$

$P_{\text{HPF}} = \dfrac{1}{2}P_{\max} = \dfrac{1}{2}(42.58\ \mu\text{W}) = \mathbf{21.29\ \mu W}$

2.

a. $BW = f_s/Q_s = 6000\text{ Hz}/15 = \mathbf{400\ Hz}$

b. $f_2 = f_s + \dfrac{BW}{2} = 6000\text{ Hz} + 200\text{ Hz} = \mathbf{6200\ Hz}$

$f_1 = f_s - \dfrac{BW}{2} = 6000\text{ Hz} - 200\text{ Hz} = \mathbf{5800\ Hz}$

c. $Q_s = \frac{X_L}{R} \Rightarrow X_L = Q_s R = (15)(3\ \Omega) = \mathbf{45\ \Omega} = X_C$

d. $P_{HPF} = \frac{1}{2} P_{max} = \frac{1}{2}(I^2 R) = \frac{1}{2}(0.5\ \text{A})^2\, 3\Omega = \mathbf{375\ mW}$

3.

$$I_M = \frac{E}{R} \Rightarrow R = \frac{E}{I_M} = \frac{5\ \text{V}}{500\ \text{mA}} = \mathbf{10\ \Omega}$$

$$BW = f_s/Q_s \Rightarrow Q_s = f_s/BW = 8400\ \text{Hz}/120\ \text{Hz} = 70$$

$$Q_s = \frac{X_L}{R} \Rightarrow X_L = Q_s R = (70)(10\ \Omega) = \mathbf{700\ \Omega}$$

$$X_C = X_L = \mathbf{700\ \Omega}$$

$$L = \frac{X_L}{2\pi f} = \frac{700\ \Omega}{(2\pi)(8.4\ \text{kHz})} = \mathbf{13.26\ mH}$$

$$C = \frac{1}{2\pi f X_C} = \frac{1}{(2\pi)(8.4\ \text{kHz})(0.7\ \text{k}\Omega)} = \mathbf{27.07\ nF}$$

$$f_2 = f_s + BW/2 = 8400\ \text{Hz} + 120\ \text{Hz}/2 = \mathbf{8.46\ kHz}$$

$$f_1 = f_s - BW/2 = 8400\ \text{Hz} - 60\ \text{Hz} = \mathbf{8.34\ kHz}$$

4.

a. $Q_\ell = \frac{X_L}{R_\ell}$

$$R_\ell = \frac{X_L}{Q_\ell} = \frac{2\pi f L}{Q_\ell} = \frac{2\pi(1\ \text{MHz})(100\ \mu\text{H})}{12.5} = 50.27\ \Omega$$

$$\frac{f_2 - f_1}{f_s} = \frac{1}{Q_s} = 0.2$$

$$Q_s = \frac{1}{0.2} = 5 = \frac{X_L}{R} = \frac{2\pi f L}{R} = \frac{2\pi(1\ \text{MHz})(100\ \mu\text{H})}{R} = \frac{628.32\ \Omega}{R}$$

$$R = \frac{628.32\ \Omega}{5} = 125.66$$

$$R = R_d + R_\ell$$

$$125.66\ \Omega = R_d + 50.27\ \Omega$$

and $R_d = 125.66\ \Omega - 50.27\ \Omega = \mathbf{75.39\ \Omega}$

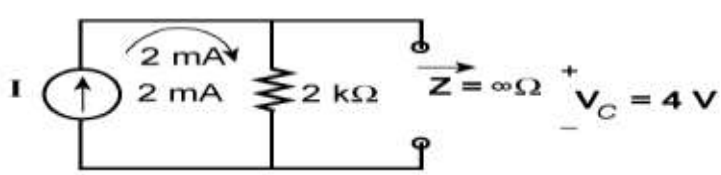
c. $X_C = \frac{1}{2\pi f C} = X_L$

$$C = \frac{1}{2\pi f X_C} = \frac{1}{2\pi(1\ \text{MHz})(628.32\ \Omega)} = \mathbf{253.3\ pF}$$

5.

a. $f_p = \frac{1}{2\pi\sqrt{LC}} = \frac{2}{2\pi\sqrt{(0.1\ \text{mH})(10\ \text{nF})}} = \mathbf{159.16\ kHz}$

b.



c. $I_L = \frac{V_L}{X_L} = \frac{4\text{ V}}{2\pi f_p L} = \frac{4\text{ V}}{100\ \Omega} = \mathbf{40\ mA}$

$I_C = \frac{V_L}{X_C} = \frac{4\text{ V}}{1/2\pi f_p C} = \frac{4\text{ V}}{100\ \Omega} = \mathbf{40\ mA}$

d. $Q_p = \frac{R_s}{X_{L_p}} = \frac{2\text{ k}\Omega}{2\pi f_p L} = \frac{2\text{ k}\Omega}{100\ \Omega} = \mathbf{20}$

6.

a. $Q_\ell = \frac{X_L}{R_L} = \frac{100\ \Omega}{20\ \Omega} = 5 \leq 10$

$\therefore \frac{X_L}{R_\ell^2 + X_L^2} = \frac{1}{X_C} \Rightarrow X_C = \frac{R_\ell^2 + X_\ell^2}{X_L} = \frac{(20\ \Omega)^2 + (100\ \Omega)^2}{100\ \Omega} = \mathbf{104\ \Omega}$

b. $Z_T = R_s \parallel R_p = R_s \parallel \frac{R_\ell^2 + X_L^2}{R_\ell} = 1000\ \Omega \parallel \frac{10{,}400\ \Omega}{20} = \mathbf{342.11\ \Omega}$

c. $\mathbf{E} = \mathbf{I}\mathbf{Z}_{T_p} = (5\text{ mA}\ \angle 0°)(342.11\ \Omega\ \angle 0°) = 1.711\text{ V}\ \angle 0$

$\mathbf{I}_C = \frac{\mathbf{E}}{X_C\ \angle -90°} = \frac{1.711\text{ V}\ \angle 0°}{104\ \Omega\ \angle -90°} = \mathbf{16.45\ mA\ \angle 90°}$

$\mathbf{Z}_L = 20\ \Omega + j100\ \Omega = 101.98\ \Omega\ \angle 78.69°$

$\mathbf{I}_L = \frac{\mathbf{E}}{\mathbf{Z}_L} = \frac{1.711\text{ V}\ \angle 0°}{101.98\ \Omega\ \angle 78.69°} = \mathbf{16.78\ mA\ \angle -78.69°}$

d. $L = \frac{X_L}{2\pi f} = \frac{100\ \Omega}{2\pi(20\text{ kHz})} = \mathbf{795.77\ \mu H}$

$C = \frac{1}{2\pi f X_C} = \frac{1}{2\pi(20\text{ kHz})(104\ \Omega)} = \mathbf{76.52\ nF}$

e. $Q_p = \frac{R}{X_C} = \frac{342.11\ \Omega}{104\ \Omega} = \mathbf{3.29}$

$BW = f_p/Q_p = 20{,}000\text{ Hz}/3.29 = \mathbf{6079.03\ Hz}$

7.

a. Ratio of X_C to R_ℓ suggests high Q system.

$\therefore X_L = \mathbf{400\ \Omega} = X_C$

b. $Q_\ell = \frac{X_L}{R_\ell} = \frac{400\ \Omega}{8\ \Omega} = \mathbf{50}$

c. $Q_p = \frac{R}{X_L} = \frac{R_s \parallel R_p}{X_L} = \frac{R_s \parallel Q_\ell^2 R_\ell}{X_L} = \frac{20\text{ k}\Omega \parallel (50)^2\ 8\ \Omega}{400\ \Omega} = \frac{10\text{ k}\Omega}{400\ \Omega} = \mathbf{25}$

$BW = \frac{f_p}{Q_p} \Rightarrow f_p = Q_p BW = (25)(1000\text{ Hz}) = \mathbf{25\ kHz}$

d. $V_{C_{max}} = IZ_{T_p} = (0.1\text{ mA})(10\text{ k}\Omega) = \mathbf{1\text{ V}}$

e. $f_2 = f_p + BW/2 = 25\text{ kHz} + \dfrac{1\text{ kHz}}{2} = \mathbf{25.5\text{ kHz}}$

$f_1 = f_p - BW/2 = 25\text{ kHz} - \dfrac{1\text{ kHz}}{2} = \mathbf{24.5\text{ kHz}}$

8.

$$V_{C_{max}} = IZ_{T_p} \Rightarrow Z_{T_p} = \frac{V_{C_{max}}}{I} = \frac{1.8\text{ V}}{0.2\text{ mA}} = 9\text{ k}\Omega$$

$$Q_p = \frac{R}{X_L} = \frac{R_s \parallel R_p}{X_L} = \frac{R_p}{X_L} \Rightarrow X_L = \frac{R_p}{Q_p} = \frac{9\text{ k}\Omega}{30} = \mathbf{300\ \Omega} = X_C$$

$$BW = \frac{f_p}{Q_p} \Rightarrow f_p = Q_p BW = (30)(500\text{ Hz}) = \mathbf{15\text{ kHz}}$$

$$L = \frac{X_L}{2\pi f} = \frac{300\ \Omega}{2\pi(15\text{ kHz})} = \mathbf{3.18\text{ mH}}$$

$$C = \frac{1}{2\pi f X_C} = \frac{1}{2\pi(15\text{ kHz})(300\ \Omega)} = \mathbf{35.37\text{ nF}}$$

$$Q_p = Q_\ell(R_s = \infty\ \Omega) = \frac{X_L}{R_\ell} \Rightarrow R_\ell = \frac{X_L}{Q_p} = \frac{300\ \Omega}{30} = \mathbf{10\ \Omega}$$